

CP 219 Project 1 — Weekly Meeting Minutes

Team: Feature Forge

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Week 1

Objective / Agenda:

Establish the foundational baseline and begin initial feature exploration for distinguishing Class A (human) versus Class B (machine) obfuscated token sequences.

Work Discussed or Completed:

- Implemented an initial approach combining TF-IDF and stylometric features.
- Evaluated multiple standard machine learning models to establish a baseline.
- Explored compressing the sparse TF-IDF representations using SVD, but this approach hurt overall performance.

Experiments / Results Discussed:

- LightGBM emerged as the strongest single model during initial evaluations.
- Early experiments with larger feature sets revealed a notable gap between cross-validation (CV) and Kaggle public leaderboard results.

Decisions Made:

- Retain the sparse TF-IDF representation rather than compressing it, despite the high dimensionality.

Next Steps:

Explore more advanced sequential and vocabulary-dynamics features to augment the TF-IDF base.

Week 2

Objective / Agenda:

Augment the feature set to better capture the underlying token generation distributions and finalize the core representation.

Work Discussed or Completed:

- Introduced Generative Markov features (class-conditional Markov transition log-likelihoods) to model sequential dependencies.
- Introduced a Heaps' Law vocabulary-growth feature to capture how unique token counts scale with document length.
- Developed a strict, leak-free validation logic to ensure fair evaluation of representation changes.

Experiments / Results Discussed:

- The combined feature set yielded a robust 150,027-feature representation (comprising TF-IDF, stylometric, Markov, and Heaps' Law features).

Decisions Made:

- Designate the 150,027-feature representation as the 'Champion' configuration. This will serve as the central benchmark for subsequent experiments.

Next Steps:

Attempt to prune the feature space or explore alternative vocabulary representations, and establish a rigorous statistical threshold for future feature additions.

Week 3

Objective / Agenda:

Experiment with alternative feature sets and address evaluation variance between validation folds.

Work Discussed or Completed:

- Designed and tested reduced TF-IDF representations and alternative vocabulary-dynamics features.
- Implemented and ran a five-seed validation protocol (seeds 42, 100, 2026, 777, and 12345) to establish a robust baseline performance distribution.
- Engineered and tested additional structural concepts, specifically Token Shape Statistics.

Experiments / Results Discussed:

- None of the alternative, reduced configurations managed to consistently outperform the 150,027-feature Champion configuration.
- The Shape Statistics configuration achieved a CV accuracy of 94.12%, failing to clear the baseline hurdle.

Decisions Made:

- Discard the pruning and alternative feature approaches.
- Reject Shape Statistics and establish an empirical Mean + 2-sigma hurdle of 94.38% CV accuracy. Any new feature or configuration must exceed this 94.38% threshold to be incorporated.
- Feature engineering is now effectively locked at the Champion representation.

Next Steps:

Pivot focus from feature engineering to model ensembling, prediction refinement, and pseudo-labeling.

Week 4

Objective / Agenda:

Develop stacking ensembles to push beyond single-model performance, evaluate pseudo-labeling, and finalize the project.

Work Discussed or Completed:

- Implemented the Phase B ensemble configuration (LightGBM and Logistic Regression meta-learner) and Phase 3 ensemble (standard and shallow LightGBM models).
- Designed and executed Phase 4 high-confidence pseudo-labeling.
- Designed and tested Similarity Augmentation (resulting in a 150,028-feature representation) to act as a dynamic feature.
- Conducted capacity ablation experiments and drafted the technical report.

Experiments / Results Discussed:

- Phase 4 pseudo-labeling achieved 94.58% leak-free CV accuracy (98.48% AUC), but its 95.53% LB score was below the Original Champion's 95.66%. The mechanism behind this CV-to-LB discrepancy was not isolated.
- Original Champion (150,027 features): 94.16% CV, 95.66% LB.
- Similarity-Augmented Champion (150,028 features): 94.38% CV, 95.93% LB.
- Phase B (LGBM + LogReg Stacking): 94.43% leak-free CV, 96.06% LB.
- Phase 3 (LGBM + Shallow LGBM Ensemble): 94.46% leak-free CV, 96.06% LB.
- Capacity Ablation Results: Accuracies dropped consistently under increasingly constrained hyperparameter settings.

Decisions Made:

- Reject Phase 4 pseudo-labeling for the final submission due to the observed CV-to-LB performance gap.
- Adopt Phase B and Phase 3 as the final ensemble configurations, both using the 150,028-feature similarity-augmented representation.

Next Steps:

Submit the finalized project report, codebase, and meeting minutes.

Project Outcome Summary

Final Feature Representation: A 150,028-feature configuration comprising TF-IDF n-grams, stylometrics, Generative Markov statistics, Heaps' Law, and one TF-IDF cosine-similarity feature evaluated under the fold-isolated validation protocol.

Validation Approach: Strict, leak-free 5-fold stratified cross-validation.

Final Configurations: Phase B and Phase 3 ensembles.

Final Recorded Result: 96.06% accuracy on the public leaderboard.